

PRAMIKA GARG

+91 81482 08354 ✉ pg2899@columbia.edu 🌐 Portfolio 📄 /in/micepram 🔄 /micepram 🎓 Scholar 🆔 ORCID

EDUCATION

Columbia University

Master of Science in Computer Science

Sep 2026 – Dec 2027 (Expected)

New York, United States of America

Vellore Institute of Technology

Bachelor of Technology in Computer Science & Engineering, GPA: 8.8/10 · 3.9/4.0 🏆

Sep 2020 – Jul 2024

Vellore, India

- **Merit Scholarship:** Ranked Top 10 / ~2100 for 2 consecutive years. 🏆

EXPERIENCE

Qualcomm

Software Engineer I

Nov 2025 – Jul 2026

Hyderabad, India

- Architected a scalable microservice validation platform using **Java, Docker, and Kubernetes**, re-engineering **Jenkins CI/CD** pipelines to reduce regression cycles from **7 days to 2 days** while expanding automated coverage by **80%**.
- Engineered distributed performance benchmarking infrastructure with **Locust, Python, and Kafka**, simulating high-concurrency IoT telemetry workloads that reduced infrastructure costs by **50%** while improving platform throughput **5x**.
- Owned end-to-end validation for IoT, AI/ML, and cloud platform releases, partnering with SWEs to investigate and resolve **100+ production-critical defects** before customer deployment.
- Collaborated across platform, infrastructure, and machine learning teams to **drive release readiness**, establish validation strategy during high-priority release cycles.

Qualcomm 🏆

Associate Software Engineer

Jul 2024 – Oct 2025

Hyderabad, India

- Engineered automated data verification microservices using **SpringBoot** across **AWS RDS and Cassandra**, eliminating manual database triage and accelerating fault analysis by **50%**.
- Built backend automation services in **Python** for mission-critical workflows including license activation, IP validation, and telemetry, eliminating repetitive manual validation and improving release quality across multi-tenant public APIs.
- Designed and shipped **"Auto Test Analyzer,"** an AI-driven developer tool that diffs regression execution reports and synthesizes edge-case datasets for **AI/ML** inference pipelines; recognized with Qualcomm's **Orion Award** for engineering impact.

GE Aerospace 🏆

Software Development Engineer Intern

Jan 2024 – Jun 2024

Bengaluru, India

- Engineered an end-to-end integration framework using **Playwright and JavaScript** for the Engine Change Tool across **40+ critical workflows**, validating state transitions and data consistency across diverse fleet scenarios.
- Streamlined asset transfer backend workflows by eliminating manual validation loops, achieving a **95% reduction** in data processing errors and saving **583 engineer hours annually**.

Expedia Group 🏆

Software Development Intern - Flights

May 2023 – Jul 2023

Gurgaon, India

- Designed and built a modular, feature-flagged configuration engine (**30+ dynamic toggles**) from scratch to drive multi-stage flight booking user flows across the Flights Line of Business.
- Scaled platform architecture across **4 Business Units**, enabling **5x faster feature development** and automating synthetic workflow verification via **GitHub Actions CI/CD**.

EPFL (Swiss Federal Institute of Technology) 🏆

Summer Research Scholar - Scalable Computing Systems Lab (SaCS Lab)

May 2022 – Jul 2022

Lausanne, Switzerland

- Selected among the **top 2%** of global applicants for a fully funded fellowship to conduct **federated machine learning experiments** focused on distributed deep learning under resource constraints.
- Designed training loops and simulation pipelines to evaluate model performance on **heterogeneous data**; implemented **lightweight pre-processing** to mitigate bandwidth bottlenecks in realistic network environments.
- Optimized aggregation strategies and communication frequency, achieving an **18% improvement in convergence speed** and reducing mean network communication overhead by **23.2%**.

SKILLS

Languages: Java, Python, JavaScript, TypeScript, Swift, SQL, C++, LaTeX

Backend & Distributed Systems: SpringBoot, FastAPI, Node.js, Kafka, RabbitMQ, REST APIs, Microservices

Cloud & DevOps: AWS, GCP, Kubernetes, Docker, EKS, Terraform, CI/CD, Linux, Git, Oracle, Splunk, Grafana

Databases & Storage: PostgreSQL, MySQL, Cassandra, AWS RDS, ChromaDB, Redis

AI, ML & Tools: TensorFlow, PyTorch, OpenCV, LangGraph, Ollama, RAG, LLMs, Federated ML, Data Analytics

Leadership & Soft Skills: JIRA, Excel, Technical Leadership, Mentorship, Cross-functional Collaboration, Agile/Scrum, Strategic Planning

Frontend & UI: React, Angular, JavaScript, jQuery, CSS, HTML

PROJECTS

Adversarial Presentation Prep Agent | Python, LangGraph, Ollama, Whisper 🏆

- Built a fully local multi-agent presentation coach using **LangGraph** and **Ollama** that conducts adversarial mock defenses by generating questions across **8 attack strategies**, simulating challenging committee-style evaluations.
- Designed a long-term memory architecture using **ChromaDB and SQLite** with episodic, semantic, document, and common-ground memories to enable personalized questioning, longitudinal feedback, and cross-session knowledge retention.
- Implemented real-time speech evaluation using **Whisper, librosa, and praat-parselmouth** to score answers across content quality, confidence, pacing, pitch variation, filler words, and delivery effectiveness using a weighted multi-dimensional rubric.

Mini Dynamo | *Java, Distributed Systems, Consistent Hashing, Replication*

- Implemented a fault-tolerant distributed key-value store inspired by **Amazon Dynamo**, supporting partitioning, replication, eventual consistency, and decentralized request routing across multiple nodes.
- Built a consistent-hashing based ring architecture with replica placement and quorum-style coordination to maintain availability and data durability during node failures and network partitions.
- Engineered recovery and synchronization mechanisms to reconcile stale replicas and restore cluster consistency after failures while preserving high write availability.

Multi-Agent Bug Triage | *Python, LangGraph, LLMs, GitHub API, RAG*

- Developed a collaborative multi-agent system that automatically analyzes GitHub issues, performs repository-aware reasoning, and routes bugs to the most relevant subsystem or owner using specialized LLM agents.
- Integrated retrieval-augmented reasoning over historical issues, repository context, and code metadata to improve issue classification accuracy and reduce repetitive manual triage effort.
- Implemented agent negotiation and consensus workflows to emulate real engineering triage processes, enabling explainable issue labeling, prioritization, and ownership recommendations.

VSCode Dead JSON Field Detector | *TypeScript, VS Code Extension API, AST Analysis*

- Built a static-analysis extension for **Visual Studio Code** that detects unused or orphaned JSON configuration fields by tracing references across large codebases and configuration graphs.
- Implemented AST-based dependency analysis and project-wide symbol resolution to identify dead configuration keys with minimal false positives while supporting large repositories efficiently.

Parsify | *Python, POSIX, Shell Parsing*

- Implemented a parser for the **POSIX.1 Shell Command Language** specification, supporting shell token recognition and lexical analysis as defined by IEEE Std 1003.1-2017.
- Designed standards-compliant parsing logic for shell grammar constructs while preserving compatibility with the unified POSIX Shell and Utilities interface.

RESEARCH PUBLICATIONS

IEEE: 4th ICUIS, 2024 iOS, ARKit, VisionOS Cited by 1 	Dec 2024
<i>Inventory Management in Augmented Reality Using Apple ARKit for VisionOS</i>	
IEEE: GCCIT, 2024 Blockchain, NFTs, Smart Contracts 	Oct 2024
<i>Leveraging BBTS and NFTs for Ethical Ticketing in Live Events</i>	
IEEE: 7th CSI-TSS, 2023 Python, MNE, Signal Processing Cited by 7 	Nov 2023
<i>Harnessing Creative Methods for EEG Feature Extraction and Modeling in Neurological Disorder Diagnoses</i>	
IEEE: 7th CSI-TSS, 2023 1st Author Python, OpenCV, Dark Channel Prior Cited by 1 	Nov 2023
<i>Enhancing Visibility: Multiresolution Dark Channel Prior for Dehazing and Fog Removal in Images</i>	
IEEE: 4th ICRTCST, 2021 1st Author Python, ML Models Cited by 1 	Feb 2022
<i>An Exhaustive Comparative Analytical Study of 15 Machine Learning Models For Automated Cardiovascular Disease</i>	
Springer: ICICIT, 2021 1st Author Python, Algorithms, Parallel Computing Cited by 2 	Jan 2022
<i>Randomised analysis of backtracking-based search algorithms in elucidating sudoku puzzles</i>	

HONORS & AWARDS

Qualcomm — Orion Award <i>Awarded for impact leading to early promotion.</i>	Aug 2025	VIT — Merit Scholarship  <i>Ranked Top 10 / 2,100+ for 2 consecutive years.</i>	2020–2022
EPFL — Summer Research Scholarship  <i>Awarded 3,600 CHF; Selected Top 2% globally.</i>	May 2022	D.E. Shaw — DESIS Educare Scholarship <i>Awarded INR 50,000 stipend.</i>	Oct 2021
Google — Generation Scholarship  <i>Awarded \$1,000; Top 180 / 20k+ (0.9% Sel.).</i>	Apr 2022	American Express — AI/ML Hackathon <i>Ranked 6th / 1,385 participants.</i>	Oct 2021
Harvard — WECODE Scholarship  <i>Selected Top 250 / 5,000+ (Selectivity: 5%).</i>	Mar 2022	SUSE — Cloud Native Scholarship  <i>Selected Top 325 / 15,000+ (Selectivity: 2%).</i>	Jun 2021

CERTIFICATIONS

O'Reilly (Packt Publishing)  <i>RAG Systems</i>	May 2026	Goldman Sachs  <i>Engineering Virtual Program</i>	Jul 2021
LinkedIn Learning  <i>Agentic AI for Developers: Concepts and Application for Enterprises</i>	May 2026	Google  <i>Deploy to Kubernetes in Google Cloud</i>	Nov 2020
LinkedIn Learning  <i>Generative AI: Introduction to Large Language Models</i>	May 2026	Google  <i>Integrate with Machine Learning APIs</i>	Nov 2020
JP Morgan Chase  <i>Software Engineering Virtual Experience</i>	Sep 2021	Google  <i>Engineer Data in Google Cloud</i>	Oct 2020